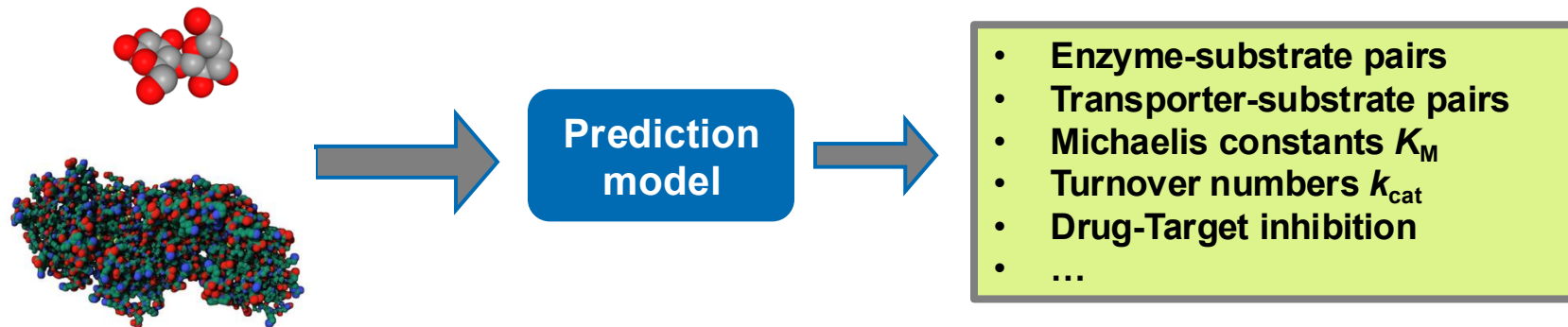
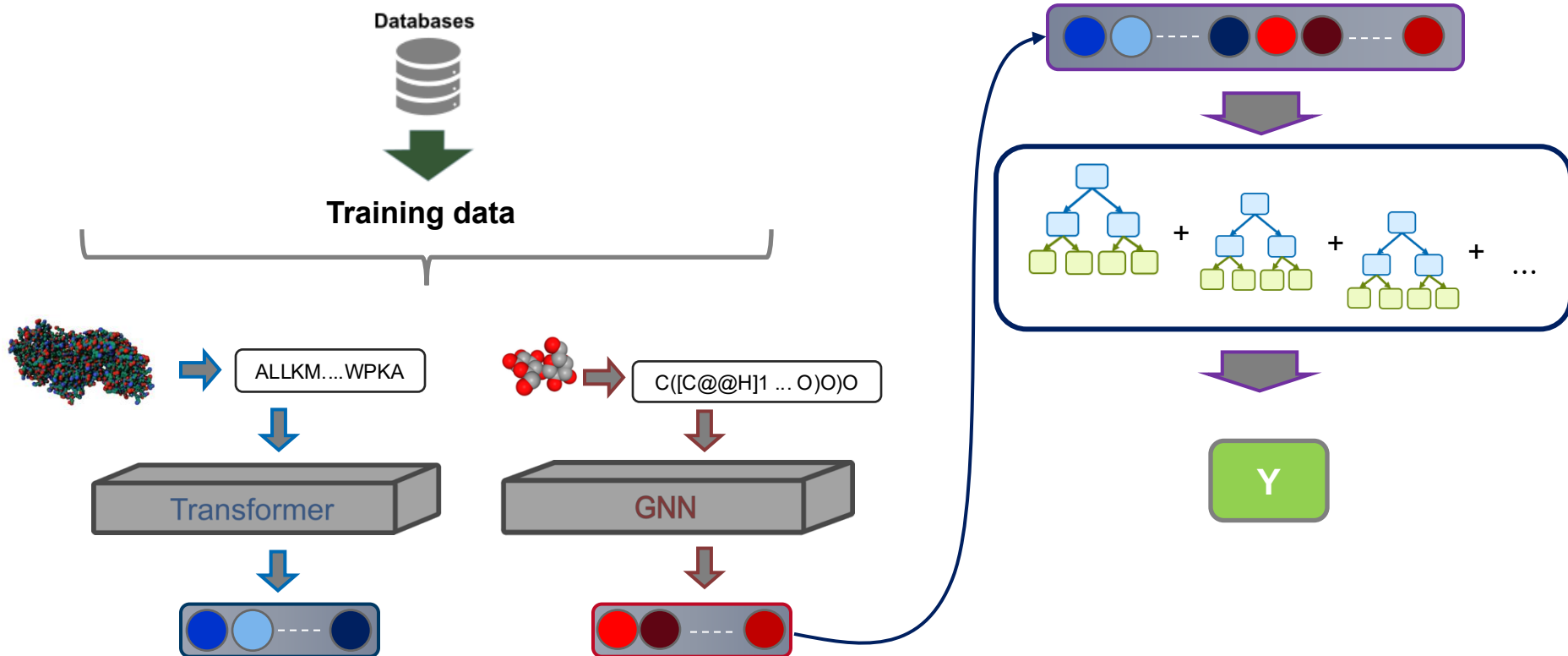


Predicting protein-small molecule interactions

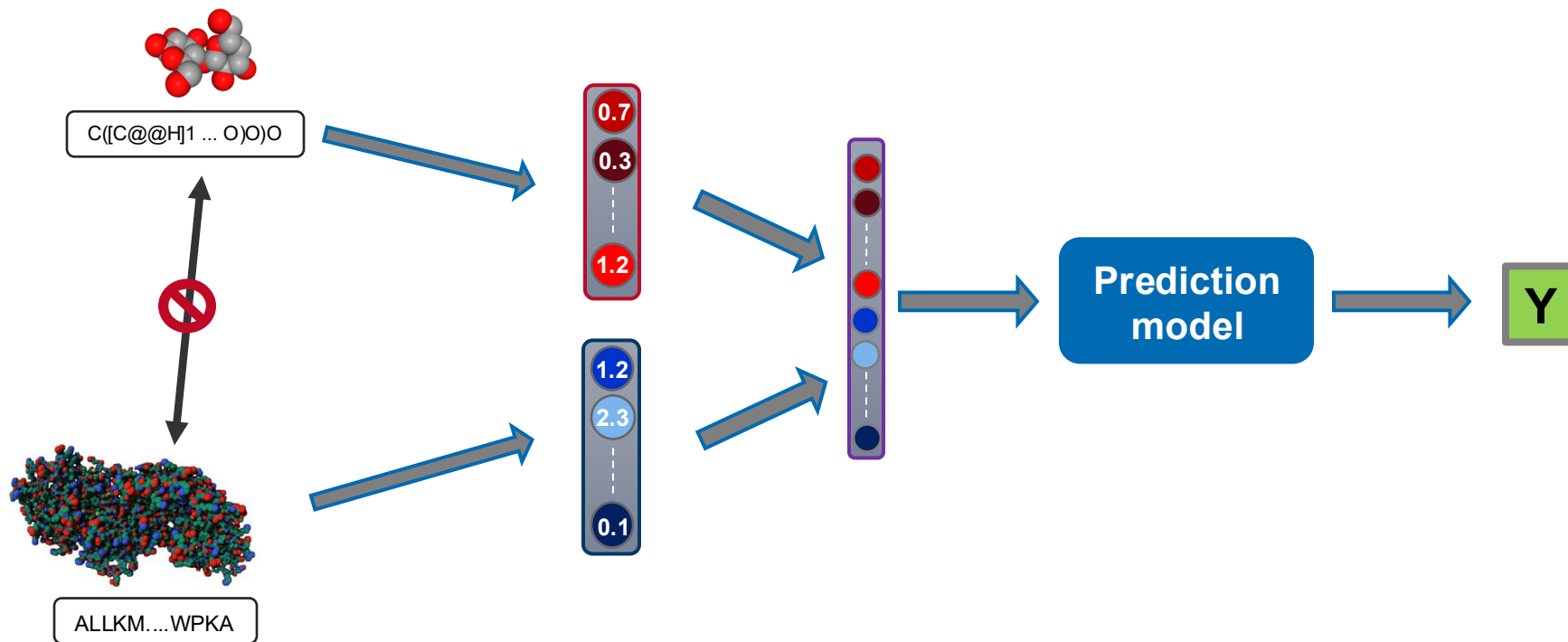
Predicting protein-small molecule interactions



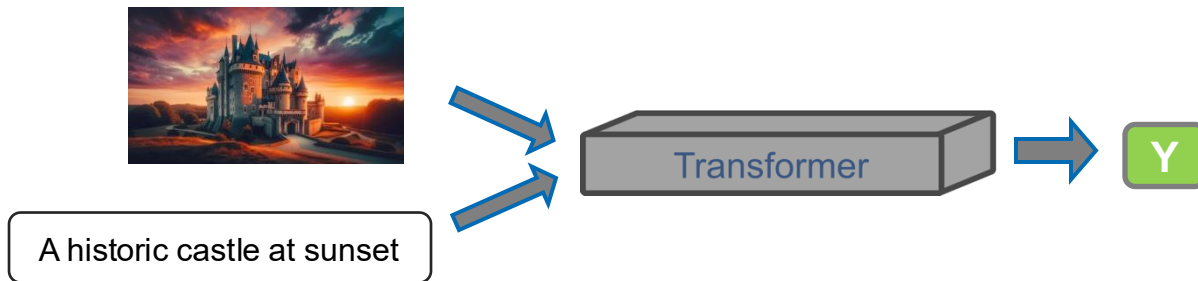
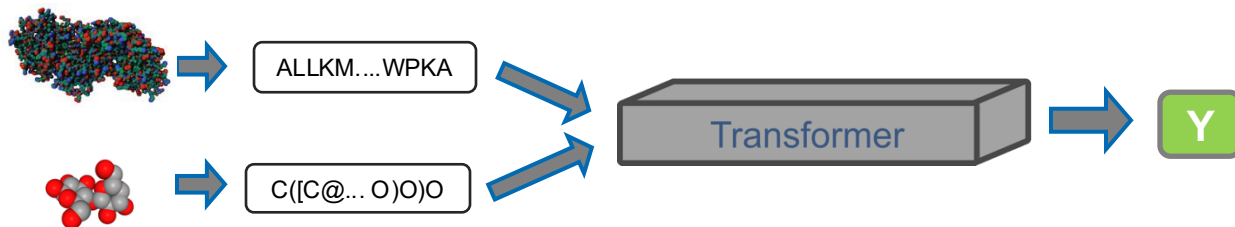
Typical Prediction Pipeline



Missing Information Exchange



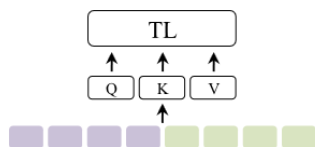
Multimodal Transformer Networks (1)



Multimodal Transformer Networks (2)

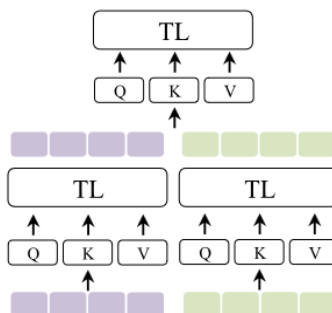
■ ■ ■ ■ = 1st input type/modality (e.g., text or protein)

■ ■ ■ ■ = 2nd input type/modality (e.g., image or small molecule)



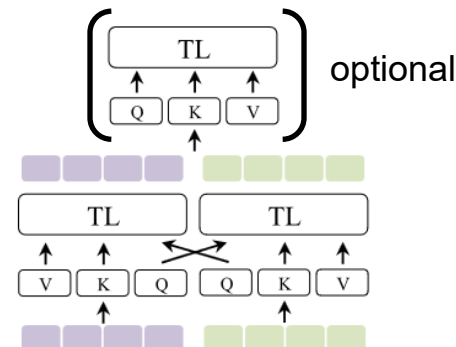
Concatenating both initial input sequences

- Generating a joint vocabulary for tokens of both input types
- Concatenated sequences is input for single Transformer



Concatenating sequence embeddings after separate processing

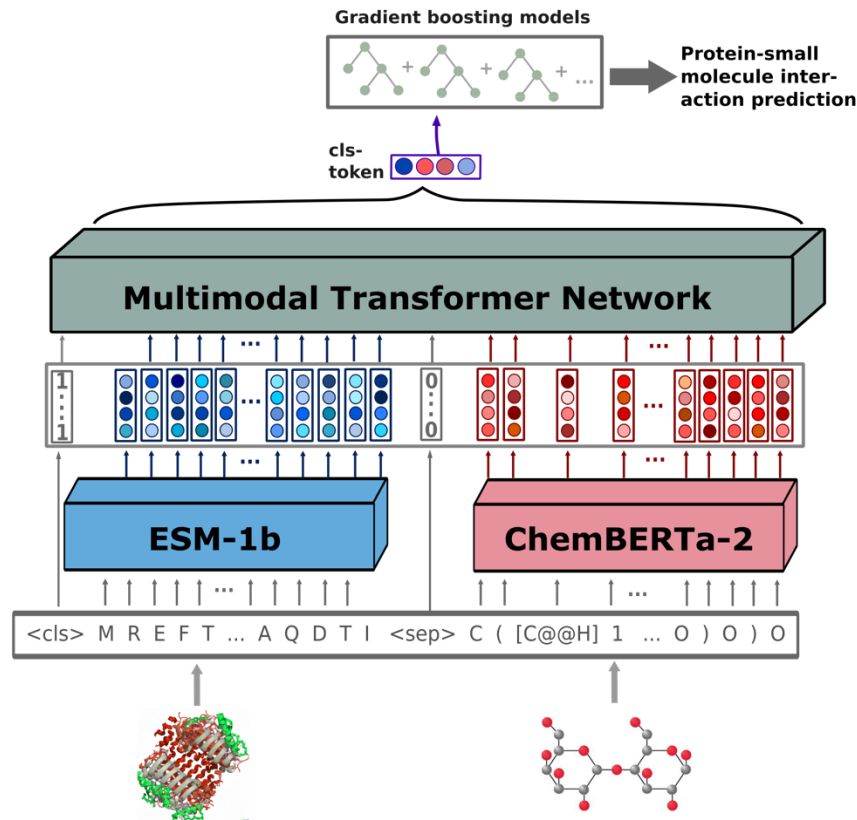
- Each input sequence is input of separate Transformer layers
- Resulting embeddings are used as the input of shared Transformer layers



Cross-attention between both sequences

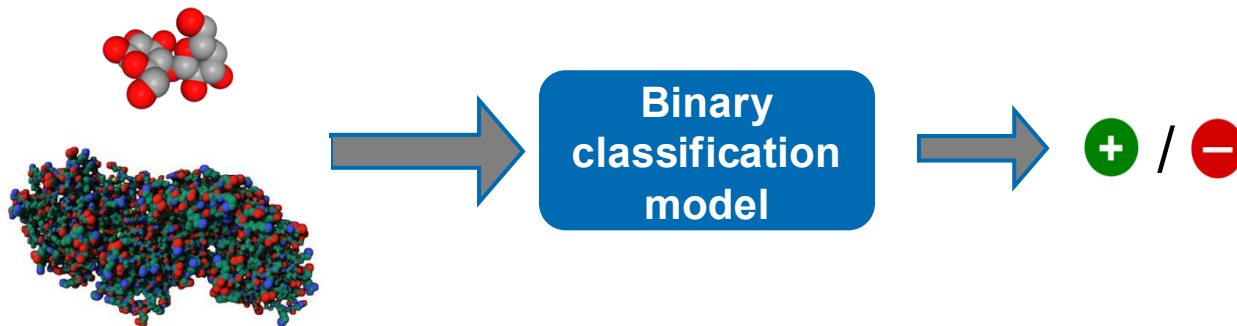
- Using embeddings from other input type to calculate attention scores
- No self-attention

ProSmith – Protein-Small Molecule Transformer



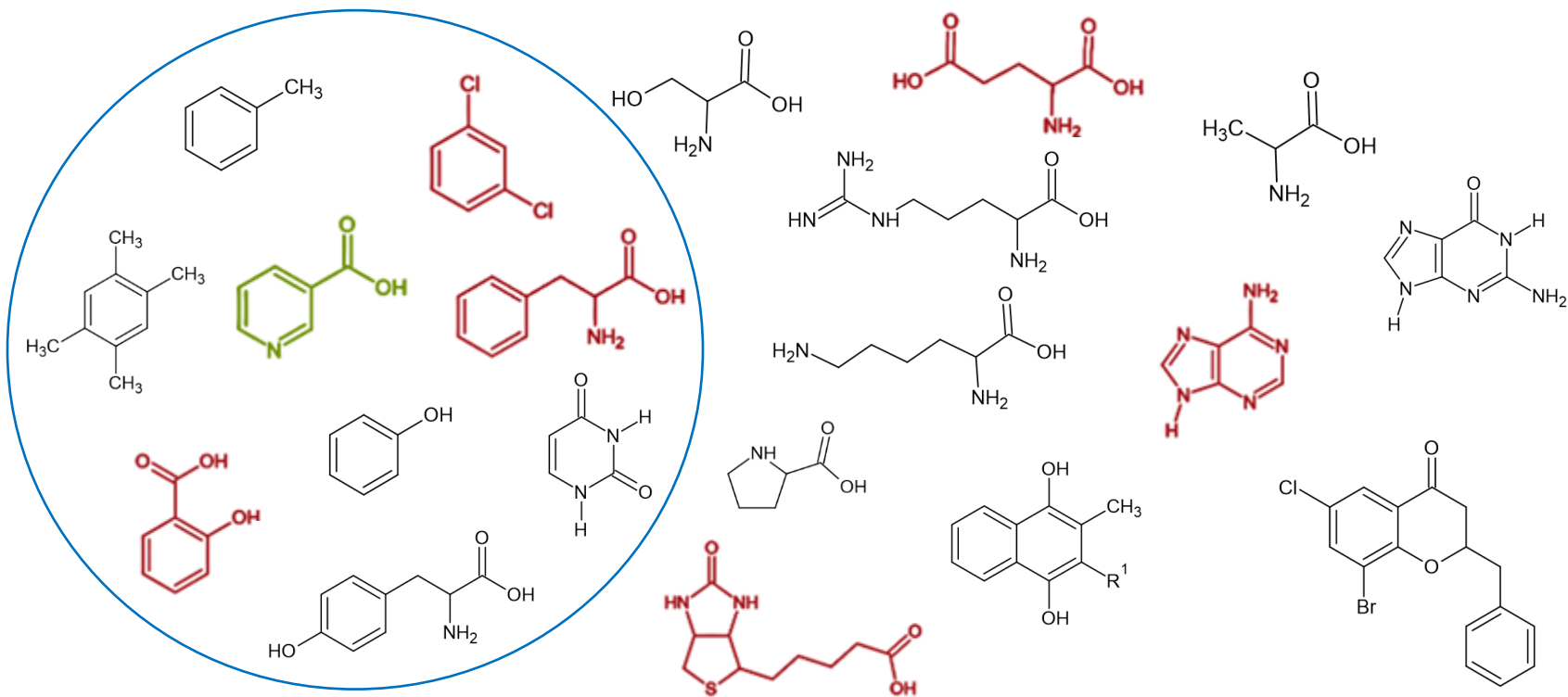
Requirement of negative data

- When training binary classification models, we need positive and negative training data



- Example: Enzyme-substrate pair prediction
 - Experimentally verified enzyme-substrate pairs are stored in protein databases
 - We also need negative data, i.e., enzyme-non substrate pairs
 - Negative data is typically not stored in large biological databases

How to add negative samples to the data set?



Summary – Sampling Negative Data Points

- When we can determine negative data points without experiments with a high likelihood, we can sample them
 - We might generate some false negative data points
- We can sample negative data points similar to the positive ones (hard negatives) to make the prediction task more difficult
 - Similarity threshold defining how similar negatives as positives are can be treated as a hyperparameter
- Negative-to-positive ratio can be treated as a hyperparameter